

Tutorial 6 — Variance of the OLS Estimator and Hypothesis Testing

Introductory Econometrics — TA Session

Duration: 50 minutes

Based on Wooldridge, Theorem 2.2, Sections 2.5 and 4.1–4.2

Block 1: Motivation — Why Isn't Unbiasedness Enough? (3 min)

The Simple Linear Regression Assumptions

We work with the model $y = \beta_0 + \beta_1 x + u$ under the following assumptions. You need to know these by name — each one will be explicitly cited in the derivations below.

Label	Name	Statement
SLR.1	Linear in Parameters	$y = \beta_0 + \beta_1 x + u$
SLR.2	Random Sampling	$\{(x_i, y_i)\}_{i=1}^n$ are i.i.d. draws
SLR.3	Sample Variation in x	$SST_x = \sum (x_i - \bar{x})^2 > 0$
SLR.4	Zero Conditional Mean	$E[u \mid x] = 0$
SLR.5	Homoskedasticity	$\text{Var}(u \mid x) = \sigma^2$ (constant)
SLR.6	Normality	$u \mid x \sim N(0, \sigma^2)$

What we proved so far: Under **SLR.1–SLR.4**, the OLS estimator is **unbiased**: $E[\hat{\beta}_1] = \beta_1$. But unbiasedness says only that $\hat{\beta}_1$ is centred at β_1 *on average* across all possible samples. It says nothing about how far any single estimate might be from the truth.

What we need now: To know how *precise* $\hat{\beta}_1$ is, we need its variance. Adding **SLR.5** allows us to derive $\text{Var}(\hat{\beta}_1)$. Adding **SLR.6** on top gives us the exact sampling distribution, which enables hypothesis tests and confidence intervals.

Theorem 2.2 (Wooldridge) — Sampling Variance of $\hat{\beta}_1$

Under **SLR.1–SLR.5**:

$$\text{Var}(\hat{\beta}_1 \mid \mathbf{x}) = \frac{\sigma^2}{SST_x} \quad \text{where} \quad SST_x = \sum_{i=1}^n (x_i - \bar{x})^2$$

The proof relies on a key representation of $\hat{\beta}_1$ that isolates the source of randomness. We derive it step by step.

Step 1: Start from the OLS formula.

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\text{SST}_x}$$

Step 2: Substitute the model (SLR.1). Since $y_i = \beta_0 + \beta_1 x_i + u_i$, taking the sample mean gives $\bar{y} = \beta_0 + \beta_1 \bar{x} + \bar{u}$. Subtracting:

$$y_i - \bar{y} = \beta_1(x_i - \bar{x}) + (u_i - \bar{u})$$

Step 3: Expand the numerator.

$$\begin{aligned} \sum (x_i - \bar{x})(y_i - \bar{y}) &= \sum (x_i - \bar{x})[\beta_1(x_i - \bar{x}) + (u_i - \bar{u})] \\ &= \beta_1 \underbrace{\sum (x_i - \bar{x})^2}_{=\text{SST}_x} + \sum (x_i - \bar{x})(u_i - \bar{u}) \end{aligned}$$

Step 4: Eliminate $\bar{u}$ from the second term.

$$\sum (x_i - \bar{x})(u_i - \bar{u}) = \sum (x_i - \bar{x})u_i - \underbrace{\bar{u} \sum (x_i - \bar{x})}_{=0} = \sum (x_i - \bar{x})u_i$$

The key fact is $\sum (x_i - \bar{x}) = 0$ (deviations from the mean always sum to zero).

Step 5: Divide by SST_x .

$$\hat{\beta}_1 = \frac{\beta_1 \cdot \text{SST}_x + \sum (x_i - \bar{x})u_i}{\text{SST}_x} = \beta_1 + \frac{\sum (x_i - \bar{x})u_i}{\text{SST}_x}$$

$$\boxed{\hat{\beta}_1 = \beta_1 + \sum_{i=1}^n w_i u_i, \quad w_i = \frac{x_i - \bar{x}}{\text{SST}_x}}$$

Interpretation: $\hat{\beta}_1$ equals the true parameter β_1 plus a weighted sum of the unobserved errors $u_1, \dots, u_n$. The weights w_i depend only on the x -values, so conditional on $\mathbf{x}$ they are constants. The *only* source of randomness in $\hat{\beta}_1$ is the errors u_i .

Block 2: Deriving and Computing $\text{Var}(\hat{\beta}_1)$ (12 min)

Question 1 (Derivation and numerical computation)

- (a) Starting from the representation $\hat{\beta}_1 = \beta_1 + \sum_{i=1}^n w_i u_i$, derive $\text{Var}(\hat{\beta}_1 | \mathbf{x}) = \sigma^2 / \text{SST}_x$.

Hint: Follow these steps — (i) Why can you drop β_1 from the variance? (ii) Expand $\text{Var}(\sum w_i u_i | \mathbf{x})$. Which assumption eliminates the covariance terms? (iii) Which assumption makes $\text{Var}(u_i | \mathbf{x})$ the same for all i ? (iv) Simplify $\sum w_i^2$.

Solution

We start from:

$$\hat{\beta}_1 = \beta_1 + \sum_{i=1}^n w_i u_i \quad \text{where} \quad w_i = \frac{x_i - \bar{x}}{\text{SST}_x}$$

Step (i): Drop the constant β_1 .

By **SLR.1** (Linear in Parameters), the model is $y = \beta_0 + \beta_1 x + u$, where β_0 and β_1 are *fixed, unknown population parameters* — they are constants, not random variables. Since adding a constant to a random variable does not change its variance (recall: $\text{Var}(a + X) = \text{Var}(X)$ for any constant a):

$$\text{Var}(\hat{\beta}_1 \mid \mathbf{x}) = \text{Var}\left(\beta_1 + \sum_{i=1}^n w_i u_i \mid \mathbf{x}\right) = \text{Var}\left(\sum_{i=1}^n w_i u_i \mid \mathbf{x}\right)$$

Note also that we condition on $\mathbf{x} = (x_1, \dots, x_n)$, so the weights $w_i = (x_i - \bar{x})/\text{SST}_x$ are treated as **constants** (they depend only on the x -values). The **only random variables** in this expression are $u_1, \dots, u_n$.

Step (ii): Expand the variance of the weighted sum.

By the general formula for the variance of a linear combination:

$$\text{Var}\left(\sum_{i=1}^n w_i u_i \mid \mathbf{x}\right) = \underbrace{\sum_{i=1}^n w_i^2 \text{Var}(u_i \mid \mathbf{x})}_{\text{variance terms}} + \underbrace{\sum_{\substack{i,j=1 \\ i \neq j}}^n w_i w_j \text{Cov}(u_i, u_j \mid \mathbf{x})}_{\text{covariance terms}}$$

Now we apply **SLR.2** (Random Sampling): the observations $\{(x_i, y_i)\}_{i=1}^n$ are drawn *independently* from the population. Since $y_i = \beta_0 + \beta_1 x_i + u_i$, the errors $u_1, \dots, u_n$ are also **independent of each other** conditional on $\mathbf{x}$.

Independence implies that $\text{Cov}(u_i, u_j \mid \mathbf{x}) = 0$ for all $i \neq j$. Therefore, **all cross-terms vanish**:

$$= \sum_{i=1}^n w_i^2 \text{Var}(u_i \mid \mathbf{x}) + 0 = \sum_{i=1}^n w_i^2 \text{Var}(u_i \mid \mathbf{x})$$

Step (iii): Apply homoskedasticity.

Now apply **SLR.5** (Homoskedasticity): the variance of the error term is the *same* for all observations, regardless of the value of x :

$$\text{Var}(u_i \mid \mathbf{x}) = \sigma^2 \quad \text{for all } i = 1, \dots, n$$

Since σ^2 does not depend on i , it can be **factored out** of the sum:

$$\sum_{i=1}^n w_i^2 \text{Var}(u_i \mid \mathbf{x}) = \sum_{i=1}^n w_i^2 \cdot \sigma^2 = \sigma^2 \sum_{i=1}^n w_i^2$$

Why SLR.5 is critical: Without homoskedasticity, each observation could have a different error variance $\text{Var}(u_i \mid x_i) = \sigma_i^2$, and we could *not* factor out a single σ^2 . The formula would become $\sum w_i^2 \sigma_i^2$, which depends on each individual σ_i^2 and is much harder to work with. This is exactly the complication that arises under **heteroskedasticity**.

Step (iv): Simplify $\sum w_i^2$.

Recall that $w_i = (x_i - \bar{x})/\text{SST}_x$. Squaring and summing:

$$\sum_{i=1}^n w_i^2 = \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{\text{SST}_x^2} = \frac{1}{\text{SST}_x^2} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{\text{SST}_x}{\text{SST}_x^2} = \frac{1}{\text{SST}_x}$$

In the third equality, we used the definition $\text{SST}_x = \sum_{i=1}^n (x_i - \bar{x})^2$. Note that **SLR.3** (Sample Variation in x) guarantees $\text{SST}_x > 0$, so this division is valid.

Combining all four steps:

$$\text{Var}(\hat{\beta}_1 | \mathbf{x}) \stackrel{(i)}{=} \text{Var}\left(\sum w_i u_i | \mathbf{x}\right) \stackrel{(ii)}{=} \sum w_i^2 \text{Var}(u_i | \mathbf{x}) \stackrel{(iii)}{=} \sigma^2 \sum w_i^2 \stackrel{(iv)}{=} \sigma^2 \cdot \frac{1}{\text{SST}_x}$$

$$\text{Var}(\hat{\beta}_1 | \mathbf{x}) = \frac{\sigma^2}{\text{SST}_x}$$

□

Summary of assumptions used:

- **SLR.1** (Linear in Parameters): β_1 is a constant, so it drops from the variance.
- **SLR.2** (Random Sampling): errors are independent, so covariance terms are zero.
- **SLR.3** (Sample Variation): $\text{SST}_x > 0$, so division is valid.
- **SLR.5** (Homoskedasticity): all $\text{Var}(u_i | \mathbf{x}) = \sigma^2$, so σ^2 factors out.

Note: **SLR.4** (Zero Conditional Mean) is *not* needed for the variance formula — it was needed for unbiasedness but not here.

- (b) Suppose $n = 5$ observations with x -values $\{1, 3, 5, 7, 9\}$ and $\sigma^2 = 10$. Compute $\bar{x}$, SST_x , $\text{Var}(\hat{\beta}_1)$, and $\text{sd}(\hat{\beta}_1)$.

Solution

Step 1: Compute $\bar{x}$.

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = \frac{1 + 3 + 5 + 7 + 9}{5} = \frac{25}{5} = 5$$

Step 2: Compute each deviation $(x_i - \bar{x})$ and its square.

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
1	$1 - 5 = -4$	$(-4)^2 = 16$
3	$3 - 5 = -2$	$(-2)^2 = 4$
5	$5 - 5 = 0$	$0^2 = 0$
7	$7 - 5 = 2$	$2^2 = 4$
9	$9 - 5 = 4$	$4^2 = 16$
		$\text{SST}_x = 40$

Step 3: Apply the formula.

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{\text{SST}_x} = \frac{10}{40} = \boxed{0.25}$$

$$\text{sd}(\hat{\beta}_1) = \sqrt{\text{Var}(\hat{\beta}_1)} = \sqrt{0.25} = \boxed{0.5}$$

Interpretation: Across repeated samples with these same x -values, the OLS slope estimate $\hat{\beta}_1$ would have a standard deviation of 0.5 — on average, our estimates will be about 0.5 units away from the true β_1 .

- (c) Based on the formula $\text{Var}(\hat{\beta}_1) = \sigma^2/\text{SST}_x$, explain what happens to the precision of $\hat{\beta}_1$ when:
- (i) the error variance σ^2 increases;
 - (ii) we add more observations that are spread out (not concentrated at $\bar{x}$);
 - (iii) we add observations that are all equal to $\bar{x}$.

Solution

The formula $\text{Var}(\hat{\beta}_1) = \sigma^2/\text{SST}_x$ has two ingredients: σ^2 in the numerator and SST_x in the denominator.

- (i) σ^2 **increases** $\implies$ **Var increases (less precise)**.

σ^2 measures the noise in the error term. More noise means the data points are more scattered around the regression line, making it harder to pin down the slope.

Numerical example: With our data ($\text{SST}_x = 40$), doubling σ^2 from 10 to 20 doubles $\text{Var}(\hat{\beta}_1)$ from 0.25 to 0.50.

- (ii) **Adding spread-out observations increases $\text{SST}_x \implies$ Var decreases (more precise)**.

Each new observation at $x_i \neq \bar{x}$ contributes $(x_i - \bar{x})^2 > 0$ to SST_x . A larger SST_x in the denominator shrinks $\text{Var}(\hat{\beta}_1)$.

Numerical example: Adding $x_6 = 11$ (far from the mean) would add $(11 - \bar{x}_{\text{new}})^2$ to SST_x , substantially reducing the variance.

Intuition: More variation in x gives us more “leverage” to estimate the slope. Imagine fitting a line through data points that are all bunched together vs. spread far apart — the spread-out data pins the slope down much more precisely.

- (iii) **Adding observations at $x = \bar{x}$ does *not* help.**

If $x_i = \bar{x}$, then $(x_i - \bar{x})^2 = 0$, so these observations contribute **nothing** to SST_x . $\text{Var}(\hat{\beta}_1)$ is unchanged.

Numerical example: Adding 100 observations all at $x = 5$ (the mean) keeps $\text{SST}_x = 40$, so $\text{Var}(\hat{\beta}_1)$ stays at 0.25.

Intuition: To estimate a slope, you need to see how y changes as x changes. If all your new data have the same x , you learn nothing new about the slope — you only learn more about the intercept.

Block 3: From σ^2 to Standard Errors (10 min)

Estimating σ^2 and the t -Distribution

The formula $\text{Var}(\hat{\beta}_1) = \sigma^2/\text{SST}_x$ involves $\sigma^2 = \text{Var}(u \mid \mathbf{x})$, which is **unknown** (we never

observe the true errors u_i). We estimate it from the **residuals** $\hat{u}_i = y_i - \hat{y}_i$:

$$\hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^n \hat{u}_i^2 = \frac{\text{SSR}}{n-2}$$

Why $n-2$ and not n ? We estimated two parameters ($\hat{\beta}_0$ and $\hat{\beta}_1$) to compute the residuals. This “uses up” 2 degrees of freedom. Dividing by $n-2$ corrects for this and makes $\hat{\sigma}^2$ an **unbiased** estimator of σ^2 : $E[\hat{\sigma}^2] = \sigma^2$.

The **standard error** of $\hat{\beta}_1$ replaces σ with $\hat{\sigma}$:

$$\text{SE}(\hat{\beta}_1) = \frac{\hat{\sigma}}{\sqrt{\text{SST}_x}}$$

In R, $\hat{\sigma}$ is reported as “**Residual standard error**” and $\text{SE}(\hat{\beta}_1)$ appears in the “**Std. Error**” column.

Adding **SLR.6** (Normality: $u \mid x \sim N(0, \sigma^2)$), the t -statistic

$$t = \frac{\hat{\beta}_1 - \beta_1}{\text{SE}(\hat{\beta}_1)} \sim t_{n-2}$$

follows a **t -distribution** with $n-2$ degrees of freedom. The t (not $N(0, 1)$) arises because $\hat{\sigma}$ in the denominator is *itself a random variable* — it varies from sample to sample, adding extra uncertainty beyond the randomness of $\hat{\beta}_1$.

Key distinction:

- If σ were **known**: $Z = \frac{\hat{\beta}_1 - \beta_1}{\sigma/\sqrt{\text{SST}_x}} \sim N(0, 1)$ (standard normal)
- Since σ is **unknown**: $t = \frac{\hat{\beta}_1 - \beta_1}{\hat{\sigma}/\sqrt{\text{SST}_x}} \sim t_{n-2}$ (t -distribution, heavier tails)

As $n \rightarrow \infty$, $\hat{\sigma} \rightarrow \sigma$ and $t_{n-2} \rightarrow N(0, 1)$, so the distinction vanishes in large samples.

The following R output will be used for Questions 2 and 3. An econometrician regressed y on x using 22 observations.

```
> model <- lm(y ~ x, data = mydata)
> summary(model)
...
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    5.2000     1.3000    4.000 0.000742 ***
x              2.4000     0.8000     A      B
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4 on 20 degrees of freedom

> confint(model)
              2.5 %    97.5 %
(Intercept) 2.488248 7.911752
x            C      D
```

Question 2 (Reading R output — variance and standard errors)

- (a) From the R output, identify $\hat{\sigma}$ (the residual standard error) and the number of observations n . Then compute $\hat{\sigma}^2$ and SST_x .

Hint: Use $SE(\hat{\beta}_1) = \hat{\sigma}/\sqrt{SST_x}$ and solve for SST_x .

Solution

Reading $\hat{\sigma}$ from the output: The line “Residual standard error: 4 on 20 degrees of freedom” tells us two things:

- $\hat{\sigma} = 4$ (the estimated standard deviation of the errors)
- Degrees of freedom = $n - 2 = 20$, which means $n = 22$ observations.

Computing $\hat{\sigma}^2$:

$$\hat{\sigma}^2 = \hat{\sigma}^2 = 4^2 = \boxed{16}$$

Computing SST_x : From the output, the “Std. Error” column for x gives $SE(\hat{\beta}_1) = 0.8$. Using the formula $SE(\hat{\beta}_1) = \hat{\sigma}/\sqrt{SST_x}$:

$$0.8 = \frac{4}{\sqrt{SST_x}}$$

Solve for $\sqrt{SST_x}$: multiply both sides by $\sqrt{SST_x}$ and divide by 0.8:

$$\sqrt{SST_x} = \frac{4}{0.8} = 5$$

Square both sides:

$$SST_x = 5^2 = \boxed{25}$$

Verification: $SE(\hat{\beta}_1) = \hat{\sigma}/\sqrt{SST_x} = 4/\sqrt{25} = 4/5 = 0.8 \checkmark$

- (b) Explain in one or two sentences why replacing σ with $\hat{\sigma}$ changes the distribution of the test statistic from standard normal to t .

Solution

When σ is **known**, $(\hat{\beta}_1 - \beta_1)/(\sigma/\sqrt{SST_x})$ is a ratio of a normal random variable (the numerator) over a *constant* (the denominator), so the ratio is exactly $N(0, 1)$. When we **replace** σ with $\hat{\sigma}$, the denominator $\hat{\sigma}/\sqrt{SST_x}$ becomes a *random variable* — it fluctuates from sample to sample because $\hat{\sigma}$ is computed from the data. The ratio is now a normal random variable divided by a random estimate of its standard deviation. This ratio has **heavier tails** than the normal (sometimes $\hat{\sigma}$ underestimates σ , inflating $|t|$), producing the t_{n-2} distribution.

Intuition: The t -distribution is “wider” than the normal to account for the fact that we *don’t know* σ exactly. As the sample size grows, $\hat{\sigma}$ becomes a better and better estimate of σ , the extra uncertainty shrinks, and $t_{n-2} \rightarrow N(0, 1)$.

- (c) Find A (the t -value) and B (the p -value) in the output. What null hypothesis does this t -statistic test?

Solution

What R reports by default: The t -value and p -value in R’s `summary()` output

always test the null hypothesis

$$H_0: \beta_1 = 0 \quad \text{vs.} \quad H_1: \beta_1 \neq 0$$

That is, R tests whether the coefficient is significantly different from **zero** (two-sided).

Finding A (t-value): The t -statistic for testing $H_0: \beta_1 = 0$ is:

$$A = t = \frac{\hat{\beta}_1 - 0}{\text{SE}(\hat{\beta}_1)} = \frac{\text{Estimate}}{\text{Std. Error}} = \frac{2.4000}{0.8000} = \boxed{3.000}$$

Finding B (p-value): The p -value is the probability of observing a t -statistic as extreme as $|t| = 3$ or more, if H_0 were true:

$$B = P(|t_{20}| > 3) = 2 \cdot P(t_{20} > 3)$$

The factor of 2 appears because this is a **two-sided** test (we count both tails). In R: `2 * pt(-3, df = 20) ≈ 0.007`.

Meaning: If β_1 were truly zero, there would be only a 0.7% chance of observing a t -statistic as large as 3 in absolute value. This is strong evidence against H_0 .

Block 4: Hypothesis Testing (13 min)

Hypothesis Testing and Confidence Intervals — The Procedure

Two-sided test of $H_0: \beta_1 = \beta_{1,0}$ vs. $H_1: \beta_1 \neq \beta_{1,0}$:

1. State the hypotheses (H_0 and H_1) and significance level α .
2. Compute the test statistic: $t = (\hat{\beta}_1 - \beta_{1,0})/\text{SE}(\hat{\beta}_1)$.
3. Find the critical value: $t_{\alpha/2, n-2}$.
4. Decision rule: Reject H_0 if $|t| > t_{\alpha/2, n-2}$.
5. State the conclusion in context.

One-sided test of $H_0: \beta_1 \leq \beta_{1,0}$ vs. $H_1: \beta_1 > \beta_{1,0}$:

- Same t -statistic.
- Reject H_0 if $t > t_{\alpha, n-2}$ (entire α in one tail).
- Since $t_{\alpha, n-2} < t_{\alpha/2, n-2}$, it is **easier to reject** in the specified direction.

$(1 - \alpha)$ **Confidence interval:** $\hat{\beta}_1 \pm t_{\alpha/2, n-2} \cdot \text{SE}(\hat{\beta}_1)$.

CI-test equivalence (two-sided only): Reject $H_0: \beta_1 = \beta_{1,0}$ at level α if and only if $\beta_{1,0}$ falls **outside** the $(1 - \alpha)$ CI.

Use the R output from Block 3. You may use the following critical values for the t_{20} distribution: $t_{0.025, 20} = 2.086$ (in R: `qt(0.975, 20)`) and $t_{0.05, 20} = 1.725$ (in R: `qt(0.95, 20)`).

Question 3 (Hypothesis testing and confidence intervals)

- (a) Using A and B from Question 2, test $H_0: \beta_1 = 0$ against $H_1: \beta_1 \neq 0$ at the 5% significance level. State your conclusion.

Solution

Step 1: State hypotheses and significance level.

- $H_0: \beta_1 = 0$ (the variable x has no effect on y)
- $H_1: \beta_1 \neq 0$ (the variable x has some effect on y)
- This is a **two-sided** test at $\alpha = 0.05$.

Step 2: Test statistic.

From Q2(c): $t = A = 3.000$.

Step 3: Critical value.

For a two-sided test at $\alpha = 0.05$ with $df = 20$: $t_{0.025, 20} = 2.086$.

Step 4: Decision.

Reject H_0 if $|t| > 2.086$. Since $|t| = 3.000 > 2.086$, we **reject** H_0 .

Alternative (using the p -value): Reject H_0 if $p < \alpha$. Since $p = B = 0.007 < 0.05 = \alpha$, we **reject** H_0 . ✓

Step 5: Conclusion.

At the 5% significance level, we reject the null hypothesis that $\beta_1 = 0$. There is statistically significant evidence that x has an effect on y .

- (b) Find C and D (the 95% confidence interval for β_1). Show your work.

Solution

Step 1: Formula.

The $(1 - \alpha) = 95\%$ confidence interval for β_1 is:

$$\hat{\beta}_1 \pm t_{\alpha/2, n-2} \cdot \text{SE}(\hat{\beta}_1)$$

Step 2: Plug in values.

- $\hat{\beta}_1 = 2.4$ (from the “Estimate” column)
- $t_{0.025, 20} = 2.086$ (critical value for 95% CI with 20 df)
- $\text{SE}(\hat{\beta}_1) = 0.8$ (from the “Std. Error” column)

Step 3: Compute the margin of error.

$$\text{Margin of error} = t_{0.025, 20} \times \text{SE}(\hat{\beta}_1) = 2.086 \times 0.8 = 1.669$$

Step 4: Compute the bounds.

$$C = \text{Lower bound} = \hat{\beta}_1 - \text{Margin of error} = 2.4 - 1.669 = \boxed{0.731}$$

$$D = \text{Upper bound} = \hat{\beta}_1 + \text{Margin of error} = 2.4 + 1.669 = \boxed{4.069}$$

Step 5: State the interval.

The 95% CI for β_1 is $[0.731, 4.069]$.

Interpretation: We are 95% confident that the true β_1 lies between 0.731 and 4.069. This means that if we repeated the sampling process many times and computed a 95% CI each time, approximately 95% of those intervals would contain the true β_1 .

Consistency check with part (a): The CI does not contain 0, which is consistent with our rejection of $H_0: \beta_1 = 0$ at 5%.

- (c) Using the confidence interval from (b), test $H_0: \beta_1 = 1$ against $H_1: \beta_1 \neq 1$ at the 5%

significance level.

Solution

Step 1: State hypotheses.

- $H_0: \beta_1 = 1$
- $H_1: \beta_1 \neq 1$
- Two-sided test at $\alpha = 0.05$.

Step 2: Apply the CI-test equivalence.

For a two-sided test, we reject $H_0: \beta_1 = \beta_{1,0}$ at level α if and only if the hypothesized value $\beta_{1,0}$ falls **outside** the $(1 - \alpha)$ CI. From part (b), the 95% CI is $[0.731, 4.069]$.

Step 3: Check whether $\beta_{1,0} = 1$ is inside or outside the CI.

Since $0.731 < 1 < 4.069$, the value $\beta_{1,0} = 1$ is **inside** the 95% CI.

Step 4: Decision.

We **fail to reject** H_0 at the 5% significance level. There is not enough evidence to conclude that $\beta_1 \neq 1$.

Verification via t -test:

$$t = \frac{\hat{\beta}_1 - 1}{\text{SE}(\hat{\beta}_1)} = \frac{2.4 - 1}{0.8} = \frac{1.4}{0.8} = 1.75$$

Decision: $|t| = 1.75 < 2.086 = t_{0.025, 20}$, so we fail to reject. ✓

Both methods (CI approach and t -test) give the same answer — they are mathematically equivalent for two-sided tests.

(d) Now test $H_0: \beta_1 \leq 1$ against $H_1: \beta_1 > 1$ at the 5% significance level.

Solution

Step 1: State hypotheses and significance level.

- $H_0: \beta_1 \leq 1$
- $H_1: \beta_1 > 1$
- This is a **one-sided** (right-tailed) test at $\alpha = 0.05$.

Step 2: Compute the test statistic.

The t -statistic uses the boundary value of H_0 (i.e., $\beta_{1,0} = 1$):

$$t = \frac{\hat{\beta}_1 - \beta_{1,0}}{\text{SE}(\hat{\beta}_1)} = \frac{2.4 - 1}{0.8} = \frac{1.4}{0.8} = 1.75$$

This is the same t -statistic as in part (c). The difference will be in the critical value.

Step 3: Find the critical value.

For a **one-sided** test at $\alpha = 0.05$ with $\text{df} = 20$: $t_{0.05, 20} = 1.725$.

Notice this is *smaller* than the two-sided critical value (2.086). The one-sided test places the entire 5% rejection probability in one tail instead of splitting it as 2.5% in each tail.

Step 4: Decision rule and decision.

For a right-tailed test: Reject H_0 if $t > t_{0.05, 20} = 1.725$.

Since $t = 1.75 > 1.725$, we **reject** H_0 .

Step 5: Conclusion.

At the 5% significance level, we reject $H_0: \beta_1 \leq 1$ in favor of $H_1: \beta_1 > 1$. There is statistically significant evidence that $\beta_1 > 1$.

(e) Parts (c) and (d) test closely related hypotheses about $\beta_1 = 1$, yet give opposite conclusions.

Explain why.

Solution

The key: different critical values.

Test	t -stat	Critical value	Exceeds CV?	Decision
Two-sided (c)	1.75	$t_{0.025,20} = 2.086$	$1.75 < 2.086$	Fail to reject
One-sided (d)	1.75	$t_{0.05,20} = 1.725$	$1.75 > 1.725$	Reject

Why the critical values differ:

- The **two-sided test** splits the 5% significance level across *both* tails: 2.5% in the left tail, 2.5% in the right tail. The critical value $t_{0.025,20} = 2.086$ is large because it must capture only 2.5% in each tail.
- The **one-sided test** places the *entire* 5% in the right tail (the direction of $H_1: \beta_1 > 1$). The critical value $t_{0.05,20} = 1.725$ is smaller because it captures 5% in one tail.

Since $t = 1.75$ falls between the two critical values ($1.725 < 1.75 < 2.086$), it is large enough to reject in the one-sided test but not large enough to reject in the two-sided test.

The trade-off:

- The **one-sided test** has more **power** (higher probability of rejecting H_0) when the true β_1 lies in the direction of H_1 (here, $\beta_1 > 1$). It achieves this by concentrating all its rejection region on one side.
- However, the one-sided test has **zero power** against departures in the *opposite* direction ($\beta_1 < 1$). If β_1 were actually much less than 1, the one-sided test would never reject.
- The **two-sided test** can detect departures in *either* direction, but at the cost of lower power in any single direction.

When to use one-sided: Only when you have strong theoretical reasons to expect the effect in a specific direction *before seeing the data*. Otherwise, use a two-sided test.

Block 5: Type I/II Errors, Size, and Power (12 min)

Type I and Type II Errors

When we perform a hypothesis test, we make a decision (reject or fail to reject H_0) based on sample data. Since the data are random, we can make **mistakes**:

	H_0 is actually true	H_0 is actually false
Reject H_0	Type I Error	Correct (Power)
Fail to reject H_0	Correct	Type II Error

- **Type I Error (false positive):** Rejecting H_0 when H_0 is *true*. We conclude there is an effect, but in reality there is none.
- **Type II Error (false negative):** Failing to reject H_0 when H_0 is *false*. We miss a real effect.

Example 1 — Courtroom trial:

- H_0 : The defendant is innocent.
- H_1 : The defendant is guilty.
- **Type I:** Convicting an innocent person. The jury finds them guilty, but they didn't commit the crime.
- **Type II:** Acquitting a guilty person. The jury finds them not guilty, but they actually did it.
- The “beyond reasonable doubt” standard sets a very low α — society considers convicting an innocent person (Type I) worse than letting a guilty person go free (Type II).

Example 2 — Drug trial:

- H_0 : $\beta_1 = 0$ (the new drug has no effect on blood pressure).
- H_1 : $\beta_1 \neq 0$ (the drug has an effect).
- **Type I:** The trial concludes the drug works (rejects H_0), but in reality it has no effect. A useless drug is marketed.
- **Type II:** The trial fails to find an effect (fails to reject H_0), but the drug actually works. A beneficial drug is never brought to market.
- $\alpha = 0.05$ means: *if the drug truly has no effect*, there is at most a 5% chance we mistakenly conclude it works.

Size and Power of a Test

Size = $P(\text{reject } H_0 \mid H_0 \text{ is true}) = P(\text{Type I Error}) = \alpha$.

The size is the probability of making a Type I error. By construction, a test at significance level α has size α : we *choose* the critical value to make this probability exactly α .

Power = $P(\text{reject } H_0 \mid H_0 \text{ is false}) = 1 - P(\text{Type II Error})$.

The power is the probability of *correctly* detecting a real effect. It depends on the **true value** β^* : the farther β^* is from β_0 (the null value), the higher the power.

What we want: Low Type I error (α small) AND high power ($1 - \text{Type II}$ large). But these are in tension: making α smaller (harder to reject) also reduces power. The standard compromise is $\alpha = 0.05$ and hoping for power ≥ 0.80 .

Question 4 (Type I/II errors and power — numerical exercise)

To simplify the calculations, assume $\sigma_{\hat{\beta}}$ is **known**, so the test statistic is $Z = (\hat{\beta} - \beta_0)/\sigma_{\hat{\beta}} \sim N(0, 1)$ under H_0 (standard normal, not t).

Consider testing $H_0: \beta = 0$ vs. $H_1: \beta \neq 0$ at $\alpha = 0.05$, with $\sigma_{\hat{\beta}} = 2$.

- (a) Define Type I and Type II error in the context of this test. Give a concrete interpretation if β measures the effect of a job training program on wages (in thousands of dollars).

Solution

Type I Error = $P(\text{reject } H_0 \mid H_0 \text{ is true})$:

We reject $H_0: \beta = 0$ even though β is actually zero. In context: we conclude that the job training program raises wages, but in reality the program has **no effect**. We waste resources implementing a useless program.

Type II Error = $P(\text{fail to reject } H_0 \mid H_0 \text{ is false})$:

We fail to reject $H_0: \beta = 0$ even though $\beta \neq 0$ (the program actually works). In context: we conclude that the job training program has no effect, and we **cancel a program that actually helps workers**. The real benefit goes undetected.

- (b) The two-sided test rejects H_0 when $|Z| > 1.96$. Show that the **size** of this test is exactly 0.05.

Solution

The size is the probability of rejecting H_0 when H_0 is true. Under $H_0: \beta = 0$:

$$Z = \frac{\hat{\beta} - 0}{\sigma_{\hat{\beta}}} = \frac{\hat{\beta}}{2} \sim N(0, 1)$$

The test rejects when $|Z| > 1.96$. So:

$$\begin{aligned} \text{Size} &= P(|Z| > 1.96 \mid H_0) \\ &= P(Z > 1.96) + P(Z < -1.96) && \text{(split into two tails)} \\ &= (1 - \Phi(1.96)) + \Phi(-1.96) && \text{(using the normal CDF)} \\ &= 0.025 + 0.025 && \text{(by symmetry of the normal)} \\ &= \boxed{0.05} \end{aligned}$$

This confirms that the critical value $z_{0.025} = 1.96$ was *chosen* to make the Type I error probability exactly $\alpha = 0.05$. That is the *definition* of a test at significance level α .

- (c) Compute the **power** of this test when the true value is $\beta^* = 6$ (the program truly raises wages by \$6,000).

Hint: Under $\beta^ = 6$, $\hat{\beta} \sim N(6, 4)$, so $Z = \hat{\beta}/2 \sim N(3, 1)$. Compute $P(|Z| > 1.96)$ using the substitution $W = Z - 3 \sim N(0, 1)$. You may use: $\Phi(1.04) = 0.851$.*

Solution

Step 1: Distribution of Z under the true β^* .

If $\beta^* = 6$, then $\hat{\beta} \sim N(\beta^*, \sigma_{\hat{\beta}}^2) = N(6, 4)$. The test statistic (computed as if H_0 were true) is:

$$Z = \frac{\hat{\beta} - 0}{2} = \frac{\hat{\beta}}{2} \sim N\left(\frac{6}{2}, 1\right) = N(3, 1)$$

So under $\beta^* = 6$, Z is not $N(0, 1)$ but rather $N(3, 1)$ — it is shifted to the right by $\delta = \beta^*/\sigma_{\hat{\beta}} = 6/2 = 3$.

Step 2: Compute power = $P(|Z| > 1.96)$ **under** $Z \sim N(3, 1)$.

Split into two tails:

$$\text{Power} = P(Z > 1.96) + P(Z < -1.96)$$

Substitute $W = Z - 3 \sim N(0, 1)$, so $Z = W + 3$:

Right tail:

$$P(Z > 1.96) = P(W + 3 > 1.96) = P(W > 1.96 - 3) = P(W > -1.04) = \Phi(1.04) = 0.851$$

Left tail:

$$P(Z < -1.96) = P(W + 3 < -1.96) = P(W < -1.96 - 3) = P(W < -4.96) = \Phi(-4.96) \approx 0.000$$

Combining:

$$\text{Power} = 0.851 + 0.000 = \boxed{0.851}$$

Interpretation: If the job training program truly raises wages by \$6,000 (with $\sigma_{\hat{\beta}} = 2$), there is an 85.1% probability that our test will correctly detect the effect and reject H_0 . This is good power.

- (d) Compute the **power** when $\beta^* = 2$ (the program raises wages by only \$2,000). Compare with part (c) and explain.

Hint: $\delta = 2/2 = 1$, so $Z \sim N(1, 1)$. You may use: $\Phi(0.96) = 0.831$.

Solution

Step 1: Distribution of Z under $\beta^* = 2$.

$$Z = \frac{\hat{\beta}}{2} \sim N\left(\frac{2}{2}, 1\right) = N(1, 1)$$

Now $\delta = 1$ (the shift from the null is smaller).

Step 2: Compute power.

With $W = Z - 1 \sim N(0, 1)$:

Right tail:

$$P(Z > 1.96) = P(W > 0.96) = 1 - \Phi(0.96) = 1 - 0.831 = 0.169$$

Left tail:

$$P(Z < -1.96) = P(W < -2.96) = \Phi(-2.96) \approx 0.002$$

Combining:

$$\text{Power} = 0.169 + 0.002 = \boxed{0.171}$$

Comparison:

True β^*	$\delta = \beta^*/\sigma_{\hat{\beta}}$	Power	Detect?
6	3	0.851	Likely yes
2	1	0.171	Likely no

Why the power is so different: Power depends on how far the true β^* is from the null value ($\beta_0 = 0$), measured in units of $\sigma_{\hat{\beta}}$. This ratio $\delta = \beta^*/\sigma_{\hat{\beta}}$ is the “signal-to-noise ratio.”

- When $\beta^* = 6$: the signal ($\delta = 3$) is strong relative to the noise. The distribution of Z is shifted far enough from zero that it almost always exceeds the critical value 1.96. The test detects the effect 85% of the time.
- When $\beta^* = 2$: the signal ($\delta = 1$) is weak. The distribution of Z overlaps heavily with the rejection region. The test misses the effect about 83% of the time (Type II error probability = $1 - 0.171 = 0.829$).

Implication: Small effects are hard to detect unless $\sigma_{\hat{\beta}}$ is small (which requires large n or large SST_x). This connects back to Block 2: increasing SST_x reduces $\text{Var}(\hat{\beta}_1) = \sigma^2/\text{SST}_x$, which reduces $\sigma_{\hat{\beta}}$, which increases δ , which increases power.

Summary of Key Formulas

Concept	Formula
Total variation in x	$\text{SST}_x = \sum_{i=1}^n (x_i - \bar{x})^2$
Variance of $\hat{\beta}_1$ (Thm. 2.2)	$\text{Var}(\hat{\beta}_1 \mid \mathbf{x}) = \sigma^2/\text{SST}_x$ [uses SLR.1–5]
Estimated error variance	$\hat{\sigma}^2 = \sum \hat{u}_i^2/(n-2)$
Standard error of $\hat{\beta}_1$	$\text{SE}(\hat{\beta}_1) = \hat{\sigma}/\sqrt{\text{SST}_x}$
t -statistic	$t = (\hat{\beta}_1 - \beta_{1,0})/\text{SE}(\hat{\beta}_1) \sim t_{n-2}$ [uses SLR.1–6]
Two-sided rejection rule	Reject if $ t > t_{\alpha/2, n-2}$
One-sided rejection rule	Reject if $t > t_{\alpha, n-2}$ (for $H_1: \beta_1 > \beta_{1,0}$)
$(1 - \alpha)$ Confidence interval	$\hat{\beta}_1 \pm t_{\alpha/2, n-2} \cdot \text{SE}(\hat{\beta}_1)$
CI–test equivalence	Reject two-sided H_0 iff $\beta_{1,0} \notin \text{CI}$
Type I Error (size)	$P(\text{reject } H_0 \mid H_0 \text{ true}) = \alpha$
Type II Error	$P(\text{fail to reject } H_0 \mid H_0 \text{ false})$
Power	$P(\text{reject } H_0 \mid H_0 \text{ false}) = 1 - P(\text{Type II})$